

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 2 - Model Comparison Presentation (BERT vs GPT vs T5 vs LLaMA)
Weightage:	10% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
Student Name:	S.Sirivally	Reg. No.:	192472371
Section / Batch:	2024	Date:	21-08-26

Instructions to Students

- **Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.**
- **Compare all four models: BERT, GPT, T5, and LLaMA.**
- Explain the **architecture and working principle** of each model.
- Compare their **pre-training objectives and capabilities**.
- Map each model to appropriate **real-world applications**.
- Identify the **strengths and limitations** of each model.
- Support the presentation using **credible research papers, official documentation, benchmarks, or other reliable sources**.
- Include at least one **comparison table or diagram**.
- Use relevant **real-world case studies or application examples**.
- Conclude by identifying the **most suitable model for the selected application** and justify the decision.
- Include proper **references/citations** in the presentation.

Code of Conduct

I __S.Sirivally / 192472371__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __S.Sirivally__

SECTION A : MODEL COMPARISON PRESENTATION (BERT VS GPT VS T5 VS LLAMA)

ARCHITECTURE-BASED COMPARISON

1. Prepare a detailed comparison of BERT, GPT, T5, and LLaMA based on architecture, training objective, input/output structure, and major capabilities.

Introduction

Natural Language Processing (NLP) has developed rapidly with the introduction of Transformer-based language models. BERT, GPT, T5, and LLaMA are important Transformer-based model families, but they use different architectures, training objectives, and input-output mechanisms.

BERT is primarily designed for language understanding, GPT is designed mainly for autoregressive language generation, T5 converts NLP problems into a unified text-to-text format, and LLaMA is a decoder-only foundation language model designed for efficient language generation and adaptation.

This report provides a detailed comparison of BERT, GPT, T5, and LLaMA based on their architecture, working principle, pre-training objective, input/output structure, capabilities, strengths, limitations, and applications.

Objective

The objectives of this comparative study are:

- To understand the architecture and working principle of BERT, GPT, T5, and LLaMA.
- To compare the pre-training objectives used by the four model families.
- To compare their input and output structures.
- To identify the major capabilities of each model.
- To analyze the strengths and limitations of the four models.
- To map each model to suitable real-world NLP applications.
- To identify the major differences between encoder-only, decoder-only, and encoder-decoder Transformer architectures.

Transformer Architecture Background

BERT, GPT, and T5 are all based on the Transformer architecture introduced for sequence-to-sequence modeling. The Transformer relies heavily on attention mechanisms instead of recurrent processing.

The major components include:

Self-attention, which allows the model to determine relationships between tokens in a sequence.

Multi-head attention, which allows the model to learn different types of relationships simultaneously.

Feed-forward neural networks, which transform the representations produced by the attention layers.

Normalization layers, which help stabilize training.

Positional information, which provides information about token order.

The original Transformer architecture contains both encoder and decoder components. Later models adapted the architecture for different purposes. BERT uses an encoder-only design, GPT uses a decoder-only design, and T5 uses both encoder and decoder components. LLaMA also belongs to the decoder-only family.

BERT Architecture and Working Principle

BERT stands for Bidirectional Encoder Representations from Transformers. It was introduced by Devlin et al. as a Transformer-based model focused primarily on language understanding. BERT uses only the encoder portion of the Transformer architecture.

BERT processes the input sequence bidirectionally. This means that when generating a contextual representation for a token, the model can use information from both the left and right sides of that token.

For example, consider:

"The bank approved the loan."

The meaning of "bank" can be understood using information from the surrounding words. BERT's bidirectional attention allows it to build contextual representations using the entire input sequence.

The simplified BERT processing flow is:

Input Text → Tokenization → Token Embeddings + Position Information → Transformer Encoder Layers → Contextual Representations → Task-Specific Output

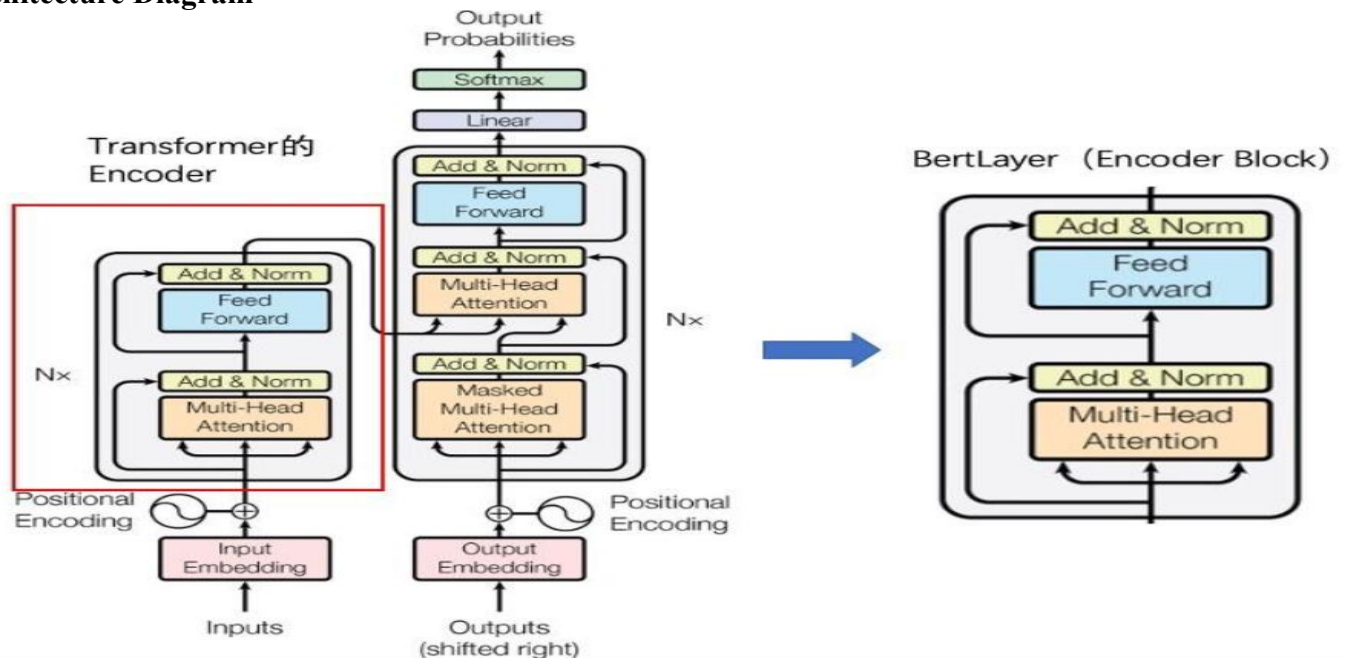
BERT is especially useful for tasks where understanding the meaning of an input sentence is more important than generating a long sequence of new text.

BERT was originally pretrained using objectives including Masked Language Modeling and Next Sentence Prediction. In Masked Language Modeling, some tokens are hidden and the model learns to predict them using surrounding context.

Major applications of BERT include:

- Text classification
- Sentiment analysis
- Named Entity Recognition
- Extractive question answering
- Sentence similarity
- Natural language inference
- Information extraction

Architecture Diagram



GPT Architecture and Working Principle

GPT stands for Generative Pre-trained Transformer. GPT models use a decoder-only Transformer architecture.

Unlike BERT, GPT processes language autoregressively. The model predicts the next token based on the tokens that have already appeared.

For example:

Input: "Artificial intelligence is"

Possible next prediction: "changing"

The process then continues:

"Artificial intelligence is changing"

The model predicts another token and continues until the requested output is generated.

The simplified GPT processing flow is:

Input Prompt → Tokenization → Token Embeddings + Positional Information → Masked Self-Attention → Transformer Decoder Layers → Next Token Prediction → Generated Output

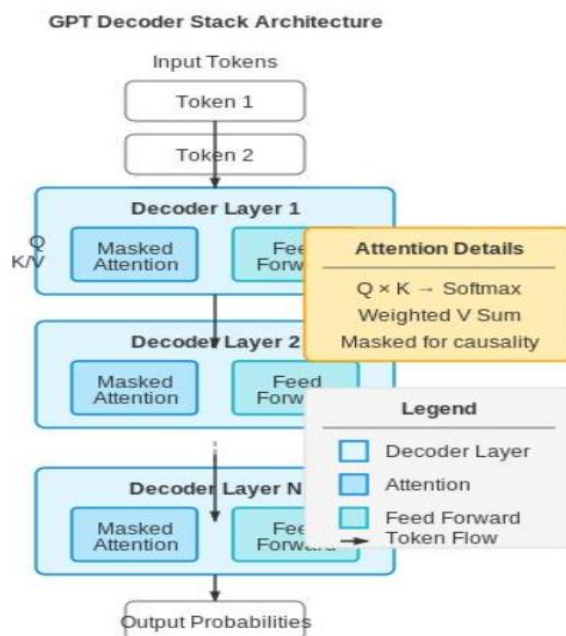
GPT uses causal or autoregressive language modeling. During training, the model learns to predict the next token using previous tokens. OpenAI's description of GPT-2, for example, explains that the model was trained to predict the next word from preceding words.

GPT models are therefore particularly suitable for:

- Text generation
- Chatbots
- Question answering
- Content generation

- Dialogue systems
- Code generation
- Creative writing
- Instruction following
- The main advantage of the decoder-only architecture is its ability to generate text naturally and continuously.

Architecture Diagram



T5 Architecture and Working Principle

T5 stands for Text-to-Text Transfer Transformer. It uses an encoder-decoder Transformer architecture. The important idea behind T5 is that different NLP tasks are represented in a common text-to-text format. Instead of designing a completely different output layer for every task, the model receives text as input and produces text as output.

For example:

Translation:

Input: "translate English to French: Hello"

Output: "Bonjour"

Summarization:

Input: "summarize: [long document]"

Output: "[short summary]"

Question answering:

Input: "question: What is AI? context: [document]"

Output: "Artificial Intelligence..."

The simplified T5 processing flow is:

Input Text $\rightarrow$ Encoder $\rightarrow$ Contextual Representation $\rightarrow$ Decoder $\rightarrow$ Generated Text

The encoder reads and represents the input, while the decoder generates the output sequence.

T5 was pretrained using a text-to-text objective involving corrupted text. The model learns to reconstruct missing or corrupted spans of text. The T5 research demonstrated the effectiveness of the unified text-to-text framework across tasks including summarization, question answering, and text classification.

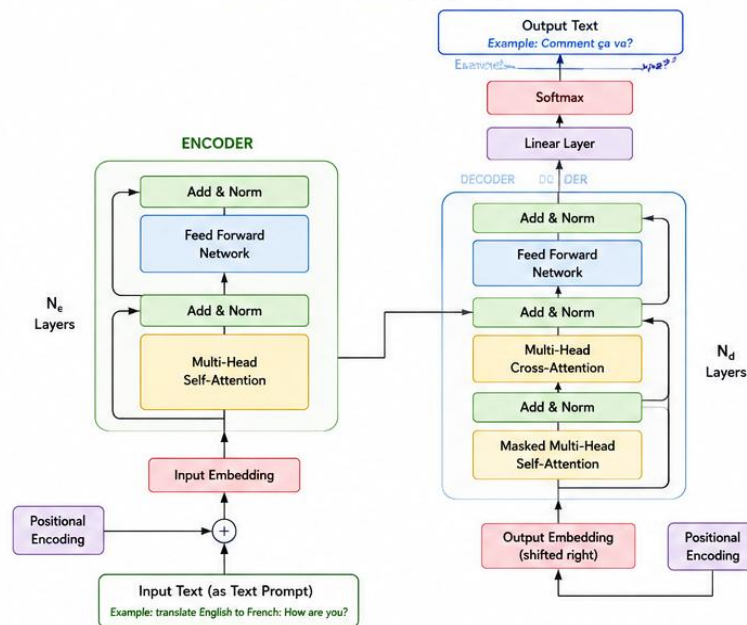
T5 is particularly suitable for:

- Machine translation
- Text summarization
- Question answering
- Text transformation
- Text classification
- Paraphrasing

Architecture Diagram

T5 ARCHITECTURE (ENCODER-DECODER)

Text-to-Text Transfer Transformer



LLaMA Architecture and Working Principle

LLaMA stands for Large Language Model Meta AI. The original LLaMA family was introduced as a collection of foundation language models ranging from 7 billion to 65 billion parameters.

LLaMA belongs to the decoder-only Transformer architecture. Therefore, it is architecturally closer to GPT than to BERT or T5.

The simplified processing flow is:

Input Prompt → Tokenization → Embedding → Transformer Decoder Blocks → Next Token Prediction → Generated Output

LLaMA uses autoregressive language modeling, where the model predicts the next token based on previously processed tokens.

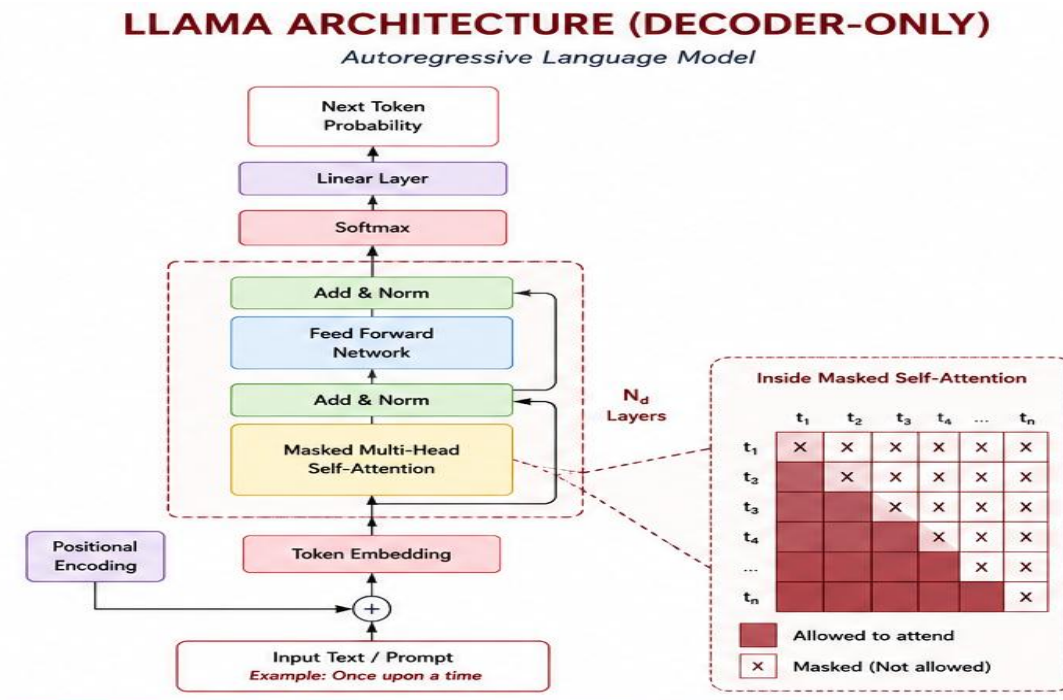
The original LLaMA research focused on training efficient foundation language models using publicly available datasets. The paper reported models at several parameter scales and demonstrated strong benchmark performance relative to larger models of that period.

LLaMA is suitable for:

- Chatbots
- Text generation
- Question answering
- Summarization
- Code-related applications
- AI assistants
- Domain-specific fine-tuning
- Local or customized LLM applications

Later Llama model families also included instruction-tuned and dialogue-oriented variants. For example, Llama 2 included pretrained and fine-tuned chat models designed for dialogue applications.

Architecture Diagram



Architecture Comparison

The four models can be classified into three major Transformer architecture types.

Model	Architecture	Main Processing Direction	Main Purpose
BERT	Encoder-only	Bidirectional	Language understanding
GPT	Decoder-only	Left-to-right / causal	Language generation
T5	Encoder-decoder	Encoder understands input, decoder generates output	Text-to-text tasks
LLaMA	Decoder-only	Left-to-right / causal	General language generation

- BERT uses only the encoder because its primary purpose is to create strong contextual representations of input text.
- GPT uses only the decoder because its primary purpose is autoregressive text generation.
- T5 uses both encoder and decoder because it first needs to understand an input sequence and then generate an output sequence.
- LLaMA uses a decoder-only design because it is primarily intended as an autoregressive language model capable of generating text from prompts.

Pre-training Objectives

- The pre-training objective determines what a model learns during its initial training stage.
- BERT uses Masked Language Modeling as a major objective. During training, some input tokens are masked and the model learns to predict them using the surrounding context. This encourages strong bidirectional language representations.
- GPT uses causal language modeling. The model predicts the next token based on previous tokens. This objective naturally supports text generation. GPT-2, for example, was trained to predict the next word using previous words in the sequence.
- T5 uses a text-to-text framework. Different NLP tasks are expressed as text input and text output. Its pre-training procedure corrupts portions of text and trains the model to reconstruct the missing information.
- LLaMA uses autoregressive language modeling and predicts the next token in a sequence. Its architecture and training approach are therefore closely related to decoder-only generative language models.

Comparison of Pre-training Objectives

Model	Pre-training Objective	Main Learning Behavior
BERT	Masked Language Modeling	Understands bidirectional context
GPT	Causal Language Modeling	Predicts next tokens
T5	Text-to-text denoising	Converts input text into output text
LLaMA	Autoregressive language modeling	Predicts next tokens

Input and Output Structure

- The input-output structure is another important difference.
- BERT generally receives a complete input sequence and produces contextual representations. A task-specific layer can then use these representations for classification or question answering.
- GPT receives a prompt and generates a continuation or response.
- T5 receives an input text and generates another text sequence.
- LLaMA receives a prompt and generates an autoregressive sequence of tokens.

Model	Input	Output
BERT	Text sequence	Contextual representation / task prediction
GPT	Prompt or previous tokens	Generated text
T5	Text formatted as a task	Generated text
LLaMA	Prompt or previous tokens	Generated text

Comparison of Contextual Understanding

- BERT has strong bidirectional contextual understanding because the encoder can attend to tokens on both sides of a target token.
- GPT and LLaMA use causal attention. A token can attend to earlier tokens but not future tokens during generation. This makes the architecture suitable for autoregressive generation.
- T5's encoder can process the input using bidirectional attention, while its decoder generates the output autoregressively.
- Therefore:
- BERT is highly suitable for understanding existing text.
- GPT is highly suitable for generating new text.
- T5 is suitable for transforming one text sequence into another.
- LLaMA is highly suitable for general-purpose autoregressive generation.

Comparison of Capabilities

Capability	BERT	GPT	T5	LLaMA
Text understanding	Excellent	Strong	Strong	Strong
Text generation	Limited	Excellent	Excellent	Excellent
Classification	Excellent	Good	Excellent	Good
Summarization	Limited without additional generation setup	Excellent	Excellent	Excellent
Translation	Limited	Possible	Excellent	Possible
Question answering	Excellent for extractive QA	Excellent for generative QA	Excellent	Excellent
Chatbot development	Limited	Excellent	Good	Excellent
Named Entity Recognition	Excellent	Possible	Good	Possible
Text transformation	Limited	Excellent	Excellent	Excellent

Strengths of BERT

- The major strengths of BERT are:
- Strong bidirectional language understanding.
- Effective contextual representations.
- Excellent performance on many classification and understanding tasks.
- Well suited for fine-tuning on specific NLP tasks.
- Suitable for tasks such as sentiment analysis, NER, and extractive question answering.

Limitations of BERT

- The major limitations of BERT are:
- It is not naturally designed for long-form text generation.
- It does not generate text autoregressively like GPT or LLaMA.
- Task-specific applications may require additional output layers or fine-tuning.
- Large BERT models can require considerable computational resources.

Strengths of GPT

- The major strengths of GPT are:
- Strong natural language generation capability.
- Suitable for interactive conversations.
- Can perform multiple tasks through prompting.

- Effective for content generation and question answering.
- Supports zero-shot and few-shot task formulation in sufficiently capable models. GPT-3 research demonstrated strong few-shot learning behavior as model scale increased.

Limitations of GPT

- The major limitations include:
- Generated responses can contain incorrect or fabricated information.
- Large models require substantial computational resources.
- Performance depends strongly on prompt design and model capabilities.
- Specialized applications may require additional grounding, fine-tuning, or retrieval systems.

Strengths of T5

- The major strengths of T5 are:
- Provides a unified text-to-text framework.
- Suitable for multiple NLP tasks.
- Strong performance in summarization and translation.
- Supports question answering and text transformation.
- Provides a consistent input-output format across different tasks.

Limitations of T5

- The major limitations include:
- Encoder-decoder processing can be more computationally expensive than simpler architectures.
- Large T5 models require significant memory and processing resources.
- For open-ended conversational applications, newer decoder-only LLMs may provide more flexible generation.
- Performance depends on model size, training data, fine-tuning, and task formulation.

Strengths of LLaMA

- The major strengths of LLaMA include:
- Decoder-only architecture suitable for text generation.
- Strong foundation for instruction-tuned and conversational applications.
- Multiple model sizes can support different deployment requirements.
- Can be adapted for specialized applications through fine-tuning.
- Its model-family approach has supported research and customized deployment.

Limitations of LLaMA

- The major limitations include:
- Larger variants require substantial computational resources.
- Generated responses can contain factual errors.
- Model performance depends on prompting and adaptation.
- Deployment of large variants may require powerful GPUs or optimized inference systems.
- Responsible deployment requires attention to safety, privacy, bias, and evaluation.

Real-World Application Mapping

BERT is appropriate for applications where understanding and classification are important.

Examples include:

- Email spam classification.
- Sentiment analysis.
- Search result ranking.

- Named Entity Recognition.
- Extractive question answering.
- Document classification.

GPT is appropriate for applications requiring flexible language generation.

Examples include:

- AI chatbots.
- Virtual assistants.
- Content generation.
- Question answering.
- Creative writing.
- Code generation.

T5 is appropriate for applications where one text sequence needs to be transformed into another.

Examples include:

- Machine translation.
- Text summarization.
- Question answering.
- Paraphrasing.
- Text classification.
- Information transformation.

LLaMA is appropriate for general-purpose generative applications.

Examples include:

- Conversational AI.
- Educational assistants.
- Local AI assistants.
- Text generation.
- Question answering.
- Domain-specific language applications.

Overall Comparison Table

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only
Main focus	Understanding	Generation	Text transformation	Generation
Attention	Bidirectional self-attention	Causal self-attention	Encoder bidirectional + decoder causal	Causal self-attention
Pre-training	Masked Language Modeling	Causal Language Modeling	Text-to-text denoising	Autoregressive Language Modeling
Input	Text sequence	Prompt	Task-formatted text	Prompt
Output	Representation/prediction	Generated text	Generated text	Generated text
Classification	Very suitable	Suitable	Suitable	Suitable
Summarization	Less natural	Very suitable	Very suitable	Very suitable
Translation	Less natural	Suitable	Very suitable	Suitable
Chatbot	Limited	Very suitable	Suitable	Very suitable

NER	Very suitable	Possible	Suitable	Possible
Main advantage	Language understanding	Flexible generation	Unified text-to-text framework	Flexible generative foundation
Main limitation	Weak direct generation	Hallucination and resource requirements	Computational cost	Resource requirements for large models

Major Differences

- The major differences between the four models can be summarized as follows:
- BERT is an encoder-only model designed mainly for understanding language.
- GPT is a decoder-only model designed mainly for generating language.
- T5 is an encoder-decoder model that treats NLP tasks as text-to-text problems.
- LLaMA is a decoder-only foundation model designed primarily for autoregressive language generation.
- BERT uses bidirectional contextual processing.
- GPT and LLaMA use causal processing for autoregressive generation.
- T5 combines bidirectional encoding with autoregressive decoding.
- BERT is particularly suitable for classification and language understanding.
- GPT and LLaMA are particularly suitable for conversational and generative applications.
- T5 is particularly suitable for sequence-to-sequence tasks such as summarization and translation.

Suitable Model for Different Tasks

Application	Most Suitable Model	Reason
Sentiment Analysis	BERT	Strong contextual representations
Named Entity Recognition	BERT	Effective token-level understanding
Extractive QA	BERT	Strong comprehension of input context
Text Generation	GPT / LLaMA	Decoder-only generation
Chatbot	GPT / LLaMA	Flexible conversational generation
Summarization	T5 / GPT / LLaMA	Strong sequence generation
Translation	T5	Encoder-decoder text transformation
Text Classification	BERT / T5	Strong task adaptation
General AI Assistant	GPT / LLaMA	Flexible generative capabilities

Findings

The comparison shows that there is no single architecture that is best for every NLP task.

- BERT is most appropriate when the main requirement is understanding and analyzing an existing text.
- GPT is most appropriate when the application requires flexible text generation and interaction.
- T5 provides a strong general framework for tasks that can be represented as input text and output text.
- LLaMA is particularly useful when a developer needs a decoder-only foundation model for generative applications and customization.

The selection of a model should therefore depend on:

- Type of NLP task.
- Required input and output format.
- Available computational resources.
- Need for text generation.
- Need for fine-tuning.
- Deployment requirements.
- Required accuracy and reliability.

Conclusion

- BERT, GPT, T5, and LLaMA are all based on the Transformer architecture, but their designs serve different purposes. BERT uses an encoder-only architecture and focuses on bidirectional language understanding. GPT uses a decoder-only architecture and focuses on autoregressive text generation. T5 combines an encoder and decoder and provides a unified text-to-text framework. LLaMA also uses a decoder-only architecture and is designed as a general-purpose foundation language model for language generation and adaptation.
- For understanding-oriented tasks such as classification, sentiment analysis, and named entity recognition, BERT is a strong choice. For flexible text generation and conversational systems, GPT and LLaMA are more appropriate. For translation, summarization, and other sequence-to-sequence tasks, T5 provides a natural architecture.
- Therefore, the most suitable model depends on the application rather than simply selecting the largest or newest model. Architecture, training objective, computational requirements, and application requirements should all be considered when selecting an NLP model.

References

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5. Touvron, H., Lavril, T., Izacard, G., et al. (2023). LLaMA: Open and Efficient Foundation Language Models. arXiv.
6. Touvron, H., Martin, L., Stone, K., et al. (2023). Llama 2: Open Foundation and Fine-Tuned Chat Models. arXiv.

REAL-WORLD CASE STUDY COMPARISON

2. A company wants to develop a general-purpose Generative AI application supporting text generation, summarization, question answering, translation, and chatbot interaction. Conduct a comprehensive comparison of BERT, GPT, T5, and LLaMA and recommend the most suitable model or combination of models.

Introduction

Generative Artificial Intelligence applications require models that can understand user instructions, process natural language, generate meaningful responses, summarize documents, answer questions, translate text, and support interactive conversations. BERT, GPT, T5, and LLaMA are all based on the

Transformer architecture, but they differ significantly in their architecture, training objectives, input-output structure, and intended applications.

BERT is an encoder-only model mainly designed for language understanding. GPT and LLaMA are decoder-only autoregressive models designed primarily for language generation. T5 uses an encoder-decoder architecture and converts different NLP problems into a unified text-to-text format. ([arXiv](#))

BERT

- BERT stands for Bidirectional Encoder Representations from Transformers. It uses an encoder-only Transformer architecture. Its self-attention mechanism allows the model to consider information from both directions when creating contextual representations.
- BERT was originally pretrained using Masked Language Modeling and Next Sentence Prediction. In Masked Language Modeling, some input tokens are hidden and the model predicts the missing tokens using their surrounding context. ([arXiv](#))

Strengths of BERT

- Provides strong contextual understanding of text.
- Suitable for classification tasks.
- Performs well for named entity recognition.
- Useful for extractive question answering.
- Can be fine-tuned for domain-specific NLP applications.
- Efficiently produces contextual representations for downstream tasks.

Limitations of BERT

- It is not primarily designed for free-form text generation.
- It cannot naturally generate long responses like decoder-based generative models.
- Additional task-specific components are generally required.
- It is less suitable as the main model for an interactive generative chatbot.

Suitable Applications

BERT is particularly suitable for:

- Sentiment analysis
- Text classification
- Named entity recognition

- Extractive question answering
- Document classification
- Search and semantic understanding

Therefore, BERT would be useful as a supporting model in a general AI system rather than as the primary generative model.

GPT

GPT is a decoder-only Transformer architecture designed for autoregressive language generation. It predicts the next token based on the tokens that have already appeared in the sequence. GPT models have demonstrated strong capabilities in text generation and few-shot learning. ([arXiv](#))

Strengths of GPT

- Excellent text generation capability.
- Suitable for conversational applications.
- Can follow natural-language instructions.
- Supports summarization and question answering.
- Can perform translation through prompting.
- Suitable for general-purpose AI assistants.
- Strong zero-shot and few-shot capabilities.

Limitations of GPT

- Large models can require significant computational resources.
- Generated answers may contain incorrect or fabricated information.
- API-based models may introduce usage costs and dependency on an external service.
- Model behavior depends strongly on prompt quality.
- Internal knowledge may not always contain the latest information.

Suitable Applications

GPT is suitable for:

- Text generation
- Chatbots
- Summarization
- Question answering
- Translation
- Content generation
- General-purpose AI assistants
- Code generation

For a company that wants a single model to perform many generative tasks, GPT is one of the strongest choices.

T5 Transformer

T5 stands for Text-to-Text Transfer Transformer. It uses an encoder-decoder Transformer architecture. The important characteristic of T5 is that it represents different NLP tasks as text-to-text problems. For

example, summarization, translation, and question answering can all be represented as an input text that produces an output text. ([Journal of Machine Learning Research](#))

Strengths of T5

- Supports multiple NLP tasks through a unified text-to-text framework.
- Well suited for summarization.
- Suitable for machine translation.
- Suitable for question answering.
- Can perform text transformation tasks.
- Can be fine-tuned for specific domains.
- Provides a clear encoder-decoder structure for input-to-output tasks.

Limitations of T5

- Large versions require considerable computational resources.
- May require fine-tuning for maximum performance on specialized applications.
- General conversational behavior may not be as strong as modern instruction-tuned decoder-only LLMs.
- Deployment requirements depend heavily on the selected T5 variant.

Suitable Applications

T5 is particularly suitable for:

- Text summarization
- Machine translation
- Question answering
- Text transformation
- Information generation
- Task-specific NLP systems

The original T5 research demonstrated its application across summarization, question answering, text classification, and other NLP tasks. ([Journal of Machine Learning Research](#))

LLaMA

LLaMA is a family of decoder-only Transformer language models. LLaMA models use autoregressive language modeling and are designed for general-purpose language generation and understanding. Llama 3, for example, supports capabilities including multilinguality, coding, reasoning, and tool usage. ([arXiv](#))

Strengths of LLaMA

- Strong general-purpose text generation.
- Suitable for conversational AI.
- Supports reasoning and instruction-following when using appropriate instruction-tuned versions.
- Can be customized and deployed in different environments.
- Suitable for local or self-hosted applications depending on model size and available hardware.
- Supports a wide range of generative AI applications.

Limitations of LLaMA

- Larger models require substantial computational resources.
- Smaller versions may provide lower quality than larger models for difficult tasks.
- Deployment requires appropriate hardware and optimization.
- Output can still contain inaccurate or hallucinated information.
- Additional safety and evaluation mechanisms may be required for production systems.

Suitable Applications

LLaMA is suitable for:

- Chatbots
- Text generation
- Question answering
- Summarization
- Coding assistants
- Educational assistants
- RAG systems
- General-purpose AI applications

Comprehensive Comparison

Feature	BERT	GPT	T5	LLaMA
Architecture	Encoder-only	Decoder-only	Encoder-decoder	Decoder-only
Main purpose	Language understanding	Language generation	Text-to-text transformation	General-purpose generation
Attention	Bidirectional	Causal	Encoder bidirectional + decoder causal	Causal
Main training objective	Masked Language Modeling	Causal Language Modeling	Text-to-text denoising	Autoregressive language modeling
Text generation	Limited	Excellent	Excellent	Excellent
Summarization	Possible with additional architecture	Excellent	Excellent	Excellent
Question answering	Excellent for extractive QA	Excellent for generative QA	Excellent	Excellent
Translation	Not the primary purpose	Possible	Excellent	Possible
Chatbot	Limited	Excellent	Possible	Excellent
Classification	Excellent	Possible	Good	Possible
Customization	Fine-tuning	Fine-tuning/prompting	Fine-tuning	Fine-tuning/parameter-efficient methods
Best role	Understanding	General generation	Text transformation	General-purpose generation

Comparison for Required Application Tasks

Text Generation

GPT and LLaMA are the strongest choices because their decoder-only architectures are designed for autoregressive text generation.

BERT is not suitable as the primary text-generation model because its architecture is designed mainly for understanding rather than autoregressive generation.

T5 can generate text because its decoder produces output sequences, but its strongest design advantage is the text-to-text framework.

Best choices: GPT or LLaMA

Summarization

T5 is particularly appropriate because summarization can naturally be represented as a text-to-text task:

Input: Long article

Output: Short summary

GPT and LLaMA can also perform summarization through prompting or instruction tuning.

BERT can help identify important information but is not naturally designed to generate a complete summary.

Best choices: T5, GPT, or LLaMA

Question Answering

BERT is highly suitable for extractive question answering, where the answer is identified from a document.

T5, GPT, and LLaMA are more suitable for generative question answering, where the system generates an answer in natural language.

Best choice depends on the requirement:

- Extractive QA → BERT
- Generative QA → GPT, T5, or LLaMA

Translation

T5's encoder-decoder architecture is naturally suitable for translation because it receives one sequence and generates another sequence.

GPT and LLaMA can also perform translation using prompting and multilingual capabilities, but translation is not their original architectural specialization.

Best choice: T5

Chatbot Interaction

GPT and LLaMA are the strongest choices for chatbot applications because decoder-only architectures naturally support conversational text generation.

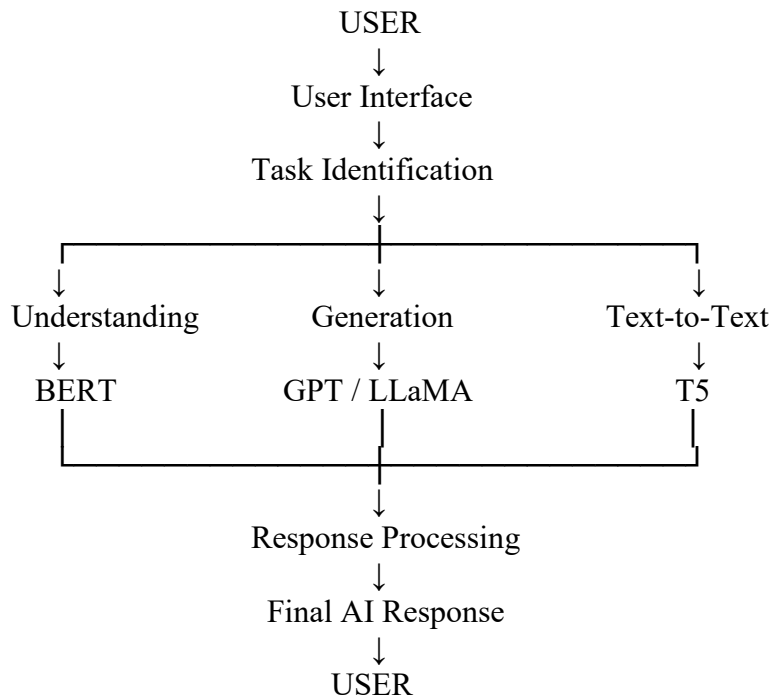
BERT can be used as an understanding component but is not ideal as the main response-generation model.

Best choices: GPT or LLaMA

Recommended System Architecture

For a general-purpose application, using one model for every task is possible, but a combination of models can provide better task specialization.

A suitable architecture is:



Recommended Model Combination

For a company developing a general-purpose Generative AI application, the recommended combination is:

Primary Model: GPT or LLaMA

GPT or LLaMA should be used as the primary model for:

- Chatbot interaction
- General text generation
- Generative question answering
- General-purpose assistance
- Instruction following

Supporting Model: T5

T5 can be used when the application requires specialized:

- Summarization
- Translation
- Text transformation
- Question answering

Supporting Model: BERT

BERT can be used for:

- Intent classification
- Sentiment analysis
- Document classification

- Named entity recognition
- Extractive question answering
- Query understanding

Recommended Choice

If the company wants simplicity and a single general-purpose model, GPT is the strongest overall choice for the five requirements because it can generate text, summarize content, answer questions, translate text, and support chatbot interaction through prompting and instruction following.

If the company prioritizes customization, deployment flexibility, and control over the model, an appropriate LLaMA-based solution can be selected.

If the company wants the best task-specialized architecture, a combination of BERT, T5, and a decoder-only generative model can be used.

The recommended production architecture would therefore be:

LLaMA/GPT as the primary generative model + T5 for specialized text-to-text tasks + BERT for language-understanding tasks.

This combination allows each architecture to perform the tasks for which it is best suited.

Advantages of the Recommended Combination

- Provides strong text-generation capabilities.
- Supports interactive chatbot applications.
- Allows specialized summarization and translation.
- Provides strong language-understanding capabilities.
- Enables task-specific optimization.
- Allows the company to select smaller models for less demanding tasks.
- Provides greater flexibility than relying entirely on one architecture.

Limitations

- Maintaining multiple models increases system complexity.
- Different models may require different hardware and software environments.
- Model switching introduces additional processing overhead.
- Multiple models require separate testing and monitoring.
- Combining models can increase deployment and maintenance costs.
- All models can produce inaccurate or hallucinated information.
- Sensitive applications require additional safety, privacy, and evaluation mechanisms.

Conclusion

BERT, GPT, T5, and LLaMA are all based on Transformer technology but are designed with different architectural objectives. BERT is an encoder-only model optimized for language understanding. T5 uses an encoder-decoder architecture and is particularly effective for text-to-text tasks such as summarization and translation. GPT and LLaMA are decoder-only models optimized for autoregressive generation and are therefore highly suitable for general-purpose generative applications. ([arXiv](#))

For the proposed general-purpose application, GPT or LLaMA should be selected as the primary generative model. T5 can be incorporated for specialized text-to-text operations, while BERT can support classification and language-understanding components. This combination provides a flexible architecture capable of supporting text generation, summarization, question answering, translation, and chatbot interaction.

References

- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. ([arXiv](#))
- Brown, T. B., et al. *Language Models are Few-Shot Learners*. ([arXiv](#))
- Raffel, C., et al. *Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer*. Journal of Machine Learning Research. ([Journal of Machine Learning Research](#))
- Grattafiori, A., et al. *The Llama 3 Herd of Models*. ([arXiv](#))

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Architectural Comparison Accuracy	30%	Accurate, in-depth comparison of architectures and use-cases	Accurate comparison with minor gaps	Superficial comparison of models	Inaccurate or confused comparison
Application Mapping	25%	Clearly maps each model to appropriate real-world applications	Reasonable mapping to applications	Weak mapping to applications	No meaningful mapping presented
Research Depth	20%	Well-researched with credible sources/benchmarks cited	Adequately researched	Limited research depth	Unresearched / opinion-based content
Delivery	15%	Confident, engaging, well-paced delivery	Clear delivery with minor pacing issues	Hesitant delivery	Unclear or unprepared delivery
Slide Design & References	10%	Well-designed slides with proper references	Adequate slides, minor reference gaps	Basic slides, poor referencing	No slides or references provided

Marks Evaluation:

Question No.	Architectural Comparison Accuracy (30%)	Application Mapping (25%)	Research Depth (20%)	Delivery (15%)	Slide Design & References (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						